2017 Year 6 Prelim II Practical Exam Preparation List

Practical Exam	Apparatus (per student)	Materials
Question 1	2x 12 cm^3 syringes 2x 6 cm^3 syringes 4x test-tubes 3x boiling tubes 1x test-tube rack 1x rubber bung with delivery tube (bung to fit boiling tube) 1x glass rod 1x stopwatch 1x permanent marker 4x paper towels 1x 500 cm^3 beaker 1x thermometer 1x distilled water bottle 1x ethanol spray	40 cm^3 7% yeast suspension, labelled **Y** 25 cm^3 0.01 mol dm^{-3} calcium hydroxide with bromothymol blue indicator, labelled **C** 25 cm^3 0.05 mol dm^{-3} hydrochloric acid, labelled **H** 20 cm^3 0.2 mol dm^{-3} glucose solution, labelled **G** 20 cm^3 0.2 mol dm^{-3} sucrose solution, labelled **S** 20 cm^3 distilled water, labelled **W** Supply of hot water and plastic cups
Question 2	1x light microscope (1 between 2 students) 3x microscope slide 3x cover slip 1x mounting needle 1x petri dish 1x forceps 1x scalpel 1x white tile	10 cm^3 1 mol dm^{-3} potassium nitrate, in drop bottle 1x 2 cm segment of rhubarb

Remarks for Q1

Chemical Y

Do not use Brewer's yeast

Y, at least 50 cm^3 of a 7.0% yeast cell suspension with glucose added **10** to **15** minutes before the candidates start Question 1 (also the starting yeast cell suspension for Question 2).
As the yeast cell suspension will froth, it should be prepared in a large container.

7.0 g of dried yeast (**for baking**) is added to 80 cm^3 of warm distilled water, stirred and made up to 100 cm^3 with warm distilled water. This should be kept at a temperature between 35 °C and 40 °C.

This is sufficient for 2 candidates.

10 to **15** minutes before the candidates start Question 1 add the glucose.
Sprinkle 20 g of glucose, a little at a time, onto the surface of the yeast cell suspension, stirring continuously. Keep warm between 35 °C and 40 °C until needed.
When ready to put into beakers for each candidate, it is suggested that the yeast cell suspension is decanted into a new beaker or container leaving the froth behind.

50 cm^3 of the yeast cell suspension should be given to each candidate in a beaker or container, labelled **Y**. The beaker or container needs to be large enough to hold at least 250 cm^3 since the yeast will froth.

Chemical H

H, at least 30 cm^3 of 0.05 $mol\,dm^{-3}$ hydrochloric acid in a beaker or container, labelled **H**.

This is sufficient for 1 candidate.

(Safety: Acid **must** be added to water. Water must **not** be added **to** acid.)

To dilute 1.00 $mol\,dm^{-3}$ hydrochloric acid, add 5 cm^3 of the acid to 95 cm^3 of distilled water in a beaker or container.

Chemical C

C, at least 70 cm^3 of 0.01 $mol\,dm^{-3}$ calcium hydroxide, in a beaker or container, labelled **C**.

This should be coloured blue by putting approximately 5 cm^3 of bromothymol blue indicator into the 70 cm^3 of **C**.

C is prepared by dissolving 0.74 g of calcium hydroxide in 500 cm^3 of distilled water, mixing well then making up to 1000 cm^3 with distilled water. This makes a solution which may remain cloudy. (If you do not have a balance which measures to 0.01 g then 0.7 g of calcium hydroxide is acceptable.)

This is sufficient for 14 candidates.

To prepare bromothymol blue indicator:

- Put 0.1 g of bromothymol blue into a beaker or container.
- Put 16 cm^3 of 0.01 $mol\,dm^{-3}$ sodium hydroxide solution into the same beaker and mix well to dissolve the bromothymol blue.
- Make up to 250 cm^3 with distilled water.

To prepare 0.01 $mol\,dm^{-3}$ sodium hydroxide solution:

- Using forceps, put 4 g of sodium hydroxide into a beaker or container with 80 cm^3 of distilled water.
- Mix well to dissolve.
- Make up to 100 cm^3 with distilled water. This is the stock solution of 1 $mol\,dm^{-3}$ sodium hydroxide solution.
- Put 1 cm^3 of this stock solution into a beaker or container and make up to 100 cm^3 with distilled water.